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HIGH-FLOW REVERSE WATER INLET CARTRIDGE SEAT

Abstract

Disclosed is a high-flow reverse water inlet cartridge seat, including: a seat body including a substrate and a cylindrical wall formed around the substrate, wherein the substrate features a top surface and a bottom surface, and the cylindrical wall protrudes from the top surface; two inlet holes and one outlet hole are triangularly configured and pierced through the top surface and the bottom surface of the substrate; two inlet enclosing frames are protruded from the top surface of the seat body; and a circular waterflow passage is formed between the two inlet enclosing frames and the cylindrical wall of the seat body to form a circumferential passage with a central passage. The circular waterflow passage features an inlet side and an outlet side, with the outlet side communicating with the outlet hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to a cartridge, in particular to an innovative high-flow reverse water inlet cartridge seat structure.

[0003] A reverse water inlet cartridge refers to a configuration in which the inlet and outlet holes of a fixed valve disk of the cartridge have a reverse relationship to the direction of flow of water into and out of the faucet.

[0004] The reverse water inlet cartridge achieves flow path redirection by coordinating the specific valve base with the fixed valve disk of a standard type to allow water to flow through the middle part between the two inlet ports after exiting the lower ceramic disk, thereby reversing the flow direction. However, limitations on the position of the cartridge base inlet port or factors such as compatibility with conventional lower ceramic disk hole positions often result in common problems of compression and reduction in flow during this process.

SUMMARY OF THE INVENTION

[0005] The main purpose of the present invention is to provide a high-flow reverse water inlet cartridge seat, aiming to innovatively address the technical problem of developing a more practical and ideal new type of water inlet cartridge seat structure.

[0006] To achieve this purpose, the present invention provides the high-flow reverse water inlet cartridge seat, comprising: a seat body including a substrate and a cylindrical wall formed around the substrate, wherein the substrate features a top surface and a bottom surface, and the cylindrical wall protrudes from the top surface; two inlet holes and one outlet hole are triangularly configured and pierced through the top surface and the bottom surface of the substrate; two inlet enclosing frames are protruded from the top surface of the seat body; and a circular waterflow passage is formed between the two inlet enclosing frames and the cylindrical wall of the seat body to form a circumferential passage with a central passage, and the circular waterflow passage features an inlet side and an outlet side, with the outlet side communicating with the outlet hole.

[0007] With this innovative structure and technical features, the present invention, as compared to the prior art, effectively increases the cross-sectional area of the flow path of the reverse water inlet cartridge through the characteristic features of the circular waterflow passage, thereby achieving improved flow rates, practical advancements, and enhanced industrial utility benefits to meet various user requirements.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is an assembled three-dimensional view of a preferred embodiment of the present invention.

[0009] FIG. 2 is an exploded three-dimensional view of a preferred embodiment of the present invention.

[0010] FIG. 3 is an exploded three-dimensional view of a preferred embodiment of the present invention in another perspective of the cartridge.

[0011] FIG. 4 is a top view of the base of a preferred embodiment of the present invention.

[0012] FIG. 5 is a bottom view of the base of a preferred embodiment of the present invention.

[0013] FIG. 6 is a vertical sectional view of a preferred embodiment of the present invention.

[0014] FIG. 7 is a schematic diagram of the flow path of a preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0015] Reference is made to FIGS. 1 through 6, which depict a preferred embodiment of a high-

flow reverse water inlet cartridge seat provided by the present invention. However, this embodiment is for illustrative purposes only and is not limited to the structure described herein. [0016] The high-flow reverse water inlet cartridge seat comprises the following components: a seat body **10**, including a substrate **11** and a cylindrical wall **12** formed around the substrate **11**, wherein the substrate **11** features a top surface **112** and a bottom surface **114**, and the cylindrical wall **12** protrudes from the top surface **112**. Two inlet holes **21** and one outlet hole **22** are triangularly configured and pierced through the top surface **112** and the bottom surface **114** of the substrate **11**. Two inlet enclosing frames **30** are protruded from the top surface **112** of the seat body **10**. A circular waterflow passage **40** is formed between the two inlet enclosing frames **30** and the cylindrical wall **12** of the seat body **10** to form a circumferential passage with a central passage **41**. The circular waterflow passage **40** features an inlet side **42** and an outlet side **43**, with the outlet side **43** communicating with the outlet hole **22**. Additionally, the two inlet enclosing frames **30** extend toward the outlet hole **22** to form extended passages **32** relatively away from the two inlet holes **21** (shown in FIG. 4 only).

[0017] As shown in FIG. 3, in this embodiment, the high-flow reverse water inlet cartridge seat is used to be mounted on the bottom of a cartridge B, and the bottom of the cartridge B includes a base **50**. Moreover, the periphery of the base **50** forms a groove **51**, and a leak prevention ring **52** is embedded in the groove **51**. The groove **51** is precisely designed to allow the cylindrical wall **12** of the seat body **10** to be nested and connected. As shown in FIG. 3, the base **50** also includes an outlet port **53**, two inlet ports **54**, and two inlet leak prevention rings **55** that separate the flow paths of the two inlet ports **54** and the outlet port **53**. The two inlet leak prevention rings **55** match the contour position of the two inlet enclosing frames **30** and tightly abut against each other. The outlet port **53** is positioned opposite the inlet side **42** of the circular waterflow passage **40**.

[0018] Within the area of each inlet leak prevention ring **55**, a baffle **56** is further formed near the outlet port **53**.

[0019] As shown in FIGS. 5 and 6, a fixed ceramic disk **60**, a movable ceramic disk **70**, a driving seat **80**, an actuating stem **90**, and a valve casing **91** are sequentially assembled above the base **50**. The fixed ceramic disk **60** includes two inlet holes **61** and an outlet hole **62**, the two inlet holes **61** corresponding to and communicating with the two inlet ports **54** of the base **50**, and the outlet hole **62** corresponding to and communicating with the outlet port **53** of the base **50**. The movable ceramic disk **70** is stacked above the fixed ceramic disk **60**, and the bottom of the movable ceramic disk **70** forms a converging passage **71** to control the communication state between the two inlet holes **61** and the outlet hole **62** of the fixed ceramic disk **60**. The actuating stem **90** is pivotally mounted in the valve casing **91** to allow a pivotal movement, and the bottom of the actuating stem **90** is provided with a connecting part **92** for driving the driving seat **80**, and the driving seat **80** is nested above the movable ceramic disk **70**.

[0020] As shown in FIG. 6, a vertical flow passage distance H1 is formed between the inlet side **42** of the circular waterflow passage **40** and the outlet hole **62** of the fixed ceramic disk **60** by the thickness of the base **50**. The advantage of this embodiment is that it can further increase the cross-sectional area of the outlet flow path so as to increase a flow rate of the water.

[0021] With the above structural configuration and technical features, when the high-flow reverse water inlet cartridge seat disclosed by the present invention is mounted on the bottom of the cartridge B in a practical application, the flow paths are as shown in FIG. 7. The cold and hot water first enter through the two inlet holes **21** of the seat body **10**, then flow sequentially through the two inlet ports **54** of the base **50**, the two inlet holes **61** of the fixed ceramic disk **60**, and then turn downward after adjusting the cold and hot water mixing ratio through the converging passage **71** at the bottom of the movable ceramic disk **70**. Thereafter, the water flows through the outlet hole **62** of the fixed ceramic disk **60**, the outlet port **53** of the base **50**, and finally through the circular waterflow passage **40** of the seat body **10**, which exits from the outlet hole **22**. Wherein, the circular waterflow passage **40**, which is formed between the two inlet enclosing frames **30** and the

cylindrical wall **12** in a circumferential passage shape, is not restricted by the narrow distance between the two inlet ports of the conventional cartridge, thereby effectively increasing the cross-sectional area of the outlet flow path, significantly improving the flow rate, and satisfying various user requirements.

[0022] It is worth mentioning that another advantage of the high-flow reverse water inlet cartridge seat disclosed by the present invention in application is that it allows for conversion and adaptation between the inlet and outlet hole positions of a general cartridge and those of a new-style faucet, because the inlet and outlet hole positions of the new-style faucet are configured opposite to those of the old-style faucet.

[0023] As shown in FIGS. **3** and **4**, one end of each inlet enclosing frame **30** is further provided with a protruding edge **31** that fits into the corresponding position of the inlet leak prevention ring **55** to form a position limiting effect.

[0024] As shown in FIGS. **3** and **5**, the bottom surface **114** of the substrate **11** forms two inlet conduits **116** communicating with the two inlet holes **21** and an outlet conduit **118** communicating with the outlet hole **22**.

[0025] Moreover, the baffle **56** corresponds to the inlet hole **21**, and the extended passage **32** corresponds to the inlet port **54**.

Claims

1. A high-flow reverse water inlet cartridge seat comprising: a seat body including a substrate and a cylindrical wall formed around the substrate, wherein the substrate features a top surface and a bottom surface, and the cylindrical wall protrudes from the top surface; two inlet holes and one outlet hole are triangularly configured and pierced through the top surface and the bottom surface of the substrate; two inlet enclosing frames are protruded from the top surface of the seat body; and a circular waterflow passage is formed between the two inlet enclosing frames and the cylindrical wall of the seat body to form a circumferential passage, and the circular waterflow passage features an inlet side and an outlet side, with the outlet side communicating with the outlet hole.
2. The high-flow reverse water inlet cartridge seat according to claim 1, wherein the high-flow reverse water inlet cartridge seat is used to be mounted on the bottom of a cartridge, and the bottom of the cartridge includes a base having an outlet port, two inlet ports, and two inlet leak prevention rings separating the flow paths of the two inlet ports and the outlet port, wherein the two inlet leak prevention rings match the contour position of the two inlet enclosing frames and tightly abut against each other, and the outlet port is positioned opposite to the inlet side of the circular waterflow passage.
3. The high-flow reverse water inlet cartridge seat according to claim 2, wherein within the area of each inlet leak prevention ring, a baffle is further formed near the outlet port.
4. The high-flow reverse water inlet cartridge seat according to claim 2, wherein a fixed ceramic disk, a movable ceramic disk, a driving seat, an actuating stem, and a valve casing are sequentially assembled above the base, wherein the fixed ceramic disk includes two inlet holes and one outlet hole, the two inlet holes corresponding to and communicating with the two inlet ports of the base, and the outlet hole corresponding to and communicating with the outlet port of the base; the movable ceramic disk is stacked above the fixed ceramic disk, and the bottom of the movable ceramic disk forms a converging passage to control the communication state between the two inlet holes and the outlet hole of the fixed ceramic disk; the actuating stem is pivotally mounted in the valve casing to allow a pivotal movement, and the bottom of the actuating stem is provided with a connecting part for driving the driving seat, and the driving seat is nested above the movable ceramic disk.
5. The high-flow reverse water inlet cartridge seat according to claim 4, wherein a vertical flow passage distance is formed between the inlet side of the circular waterflow passage and the outlet

hole of the fixed ceramic disk by the thickness of the base.

6. The high-flow reverse water inlet cartridge seat according to claim 2, wherein one end of each inlet enclosing frame is further provided with a protruding edge that fits into the corresponding position of the inlet leak prevention ring to form a position limiting effect.

7. The high-flow reverse water inlet cartridge seat according to claim 3, wherein the two inlet enclosing frames extend toward the outlet hole to form extended passages relatively away from the two inlet holes.

8. The high-flow reverse water inlet cartridge seat according to claim 7, wherein the baffle corresponds to the inlet hole, and the extended passage corresponds to the inlet port.
